

Web Programming Group Assignment (4 members per group).

Due Date: **28th November 2025**

Aunt Joy's Restaurant intends to develop a proprietary food online ordering platform to automate its sales process and directly address the growing demand for digital convenience in Mzuzu. This initiative is designed to diversify revenue by expanding its customer base beyond walk-in traffic and capturing new market segments. She projects this investment will increase sales by 25-40% within the first year and achieve a full return on investment within 12 months. Ultimately, this project is a strategic move to secure a stronger competitive position and ensure the restaurant's sustained growth

As a software developer you have been tasked to develop a web-based food ordering application for Aunt Joy restaurant. The application should meet the following functional requirements.

2.0 Functional Requirements:

2.1 Customer Interface

The system shall enable customers to:

- ✓ Browse a dynamic menu of available meals, categorized for ease of navigation.
- ✓ Utilize a search functionality to find specific meal items by name or keyword.
- ✓ Select desired meals, specify quantities, and add them to a virtual shopping cart.
- ✓ Securely submit and confirm their orders through the platform, providing necessary delivery details (e.g., delivery address within Mzuzu, contact number).

2.2 Administrator Interface

The system shall provide an administrative interface to:

- ✓ Create, read, update, and delete (CRUD) meal entries in the database, including details such as name, description, price, image, and category.
- ✓ Manage meal availability status (e.g., in stock, out of stock).
- ✓ Manager user creation and system roles.

2.3 Sales Personnel Interface

The system shall provide a dedicated interface to:

- ✓ Access a real-time list of newly placed orders with full customer and order details.

- ✓ Update the status of orders ("**Preparing**," "**Out for Delivery**," "**Delivered**") to reflect progress.

2.4 Manager Interface

The system shall provide a reporting dashboard to:

- ✓ Generate and view sales reports filtered by a specific month and year.
- ✓ Have these reports include key metrics such as total revenue, number of orders, and best-selling items.
- ✓ The system should be able to export the reports in pdf and excel formats.

3.0 Technical Specifications

- ✓ The application data shall be persisted in a normalized MySQL relational database.
- ✓ The system shall implement a secure role-based access control (RBAC) system for Customers, Administrators, Sales Staff, and Managers.
- ✓ The application must be developed as a **web-based solution** using object-oriented PHP, MySQL, HTML, CSS, and JavaScript.
- ✓ Ensure all form fields are validated before submitting data.